PRACTICAL-6

1.Aim

Interface 16x2 LCD with Raspberry Pi using 12C module to display text.

2. Prior Concepts

i) Concept of Python Programming.

ii) LCD function and use.

3. New Concepts

Hardware connection of 16 X 2 LCD with Raspi using I2C Module.

4. Hardware Requirement and PIN Connection.

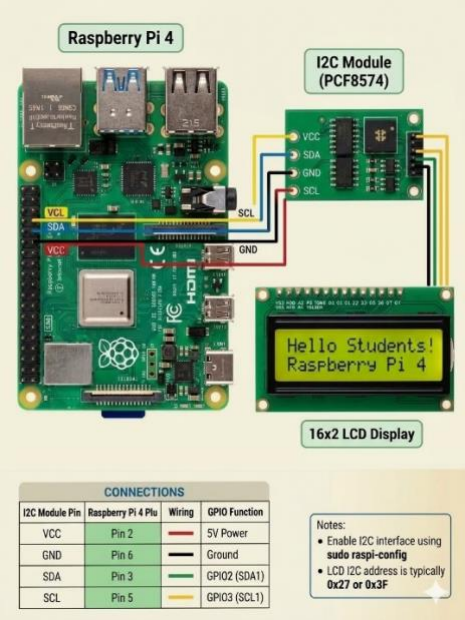
i) Raspberry Pi 4

ii) 12C LCD Module (PCF8574)

iii) Jumper Wires

Pin Connection

LCD I2C Module PIN	RPi Pin	Function
GND(Ground)	6	GND(Ground)
VCC(+5V Power)	2	5V
SDA	3	GPIO 2(SDA1)
SCL	5	GPIO 3(SCL1)



5. Procedure

i) Enable I2C Interface on Raspberry Pi by writing below code in command terminal

sudo raspi-config

ii) Navigate:

Interface Options

→ I2C

→ Enable

iii) Reboot Raspi by writing below code in command terminal

sudo reboot

iv) Install I2C Tools by writing below code in command terminal

sudo apt update

sudo apt install -y i2c-tools python3-smbus

Verify whether LCD is detected

sudo i2cdetect -y 1

Expected Output:

0123456789abcdef

20:-- -- -- -- -- -- 27 -- -- -- -- --

Note: The LCD address is typically Ox27 or Ox3F,

v)Install Install LCD Library on Raspberry Pi by writing below code in command terminal.

pip3 install RPLCD

Write the below code:

```
from RPLCD.i2c import CharLCD
from time import sleep

lcd = CharLCD(
        i2c_expander='PCF8574',
        address=0x27,
        port=1,
        cols=16,
        rows=2
)
I
cd.clear()

lcd.write_string("Hello Students!")
lcd.cursor_pos = (1, 0)
lcd.write_string("Raspberry Pi 4")

sleep(10)

lcd.clear()
```

6. Results: